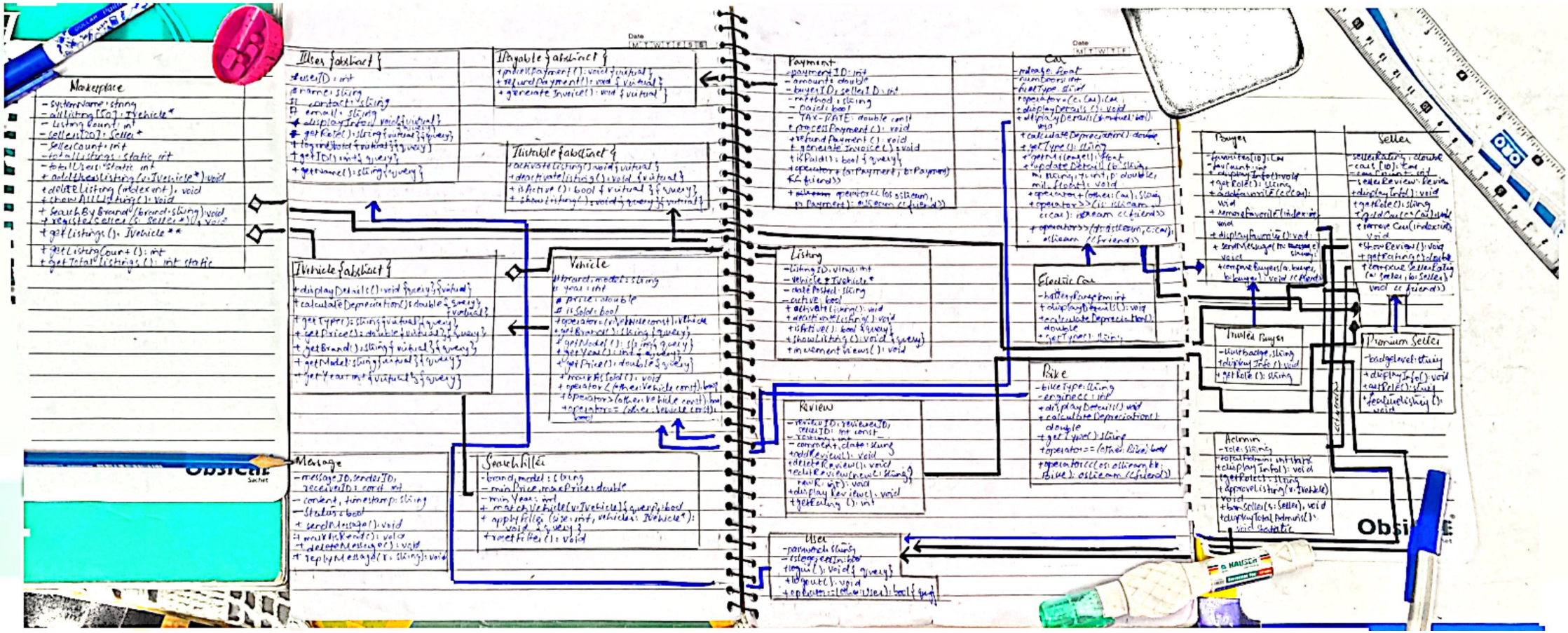
Marketplace
- systemName : string
- allListing[50] : IVehicle*
- listingCount : int
- sellers[20] : Seller*
- sellerCount : int
- totalUsers : static int
+ addNewListing(v:IVehicle*) : void
+ deleteListing(index:int) : void
+ showALLListings() : void
+ SearchByBrand(brand:string) : void
+ registerSeller(s: Seller*) : void
+ getListings() : IVehicle**
+ getListingCount() : int
+ getTotalUsers() : int static
IUser {abstract}
#userID : int
#name : string
#contact : string
#email : string
+ displayInfo() : void {virtual}
+ getRole() : string {virtual} {query}
+ toJson() : string {virtual} {query}
+ getID() : int {query}
+ getName() : string {query}
IPayable {abstract}
+ processPayment() : void {virtual}
+ refundPayment() : void {virtual}
+ generateInvoice() : void {virtual}
ITitrable {abstract}
+ deactivate(string) : void {virtual}
+ deactivateListing() : void {virtual}
+ isActive() : bool {virtual} {query}
+ show(string) : void {query} {virtual}
IVehicle {abstract}
+ displayDetails() : void {query} {virtual}
+ calculateDepreciation() : double {query} {virtual}
+ getType() : string {virtual} {query}
+ getPrice() : double {virtual} {query}
+ getBrand() : string {virtual} {query}
+ getModel() : string {virtual} {query}
+ getYear() : int {virtual} {query}
Vehicle
#brand, model : string
year : int
price : double
isSold : bool
+ operator=(v:IVehicle const) : IVehicle
+ getBrand() : string {query}
# getModel() : string {query}
# getYear() : int {query}
+ getPrice() : double {query}
+ markAsSold() : void
+ operator<(other:IVehicle const) : bool
+ operator>(other:IVehicle const) : bool
+ operator==(other:IVehicle const) : bool
Message
-messageID, senderID, receiverID : const int
- content, timestamp : string
- status : bool
+ sendMessage() : void
+ markAsRead() : void
+ deleteMessage() : void
+ replyMessage(r : string) : void
SearchFilter
-brand, model : string
-minPrice, maxPrice : double
-minYear : int
+ matchVehicle(v:IVehicle) {query} : bool
+ applyFilter(s:int, vehicles: IVehicle*) : void {query}
+ resetFilter() : void
Payment
-paymentID : int
- amount : double
- buyerID, sellerID : int
- method : string
- paid : bool
- TAX_RATE : double const
+ processPayment() : void
+ refundPayment() : void
+ generateInvoice() : void
+ isPaid() : bool {query}
+ operator+(a:Payment, b:Payment) {friend}
+ operator<<(os:ostream, p:Payment) : ostream {friend}
Listing
-listingID, views : int
-vehicle : IVehicle*
- datePosted : string
- active : bool
+ activate(string) : void
+ deactivateListing() : void
+ isActive() : bool {query}
+ showListing() : void {query}
+ incrementViews() : void
Review
-reviewID, reviewerID, sellerID : int const
- rating : int
- comment, date : string
+ addReview() : void
+ deleteReview() : void
+ editReview(newRating : int, newComment : string) : void
+ displayReview() : void
+ getRating() : int
User
-username, string
-password : string
+login() : void {query}
+ logout() : void
+ operator==(other:User) : bool {query}
Car
-mileage : float
-numDoors : int
-fuelType : string
+ operator=(c:Car) : Car
+ displayDetails() : void
+ displayDetails(b:mutable bool) : void
+ calculateDepreciation() : double
+ getType() : string
+ getMileage() : float
+ updateDetails(b:string, m:string, y:int, p:double, mil:float) : void
+ operator+(other:Car) : string
+ operator>>(is:istream, c:Car) : istream {friend}
+ operator<<(os:ostream, c:Car) : ostream {friend}
Electric Car
-batteryRange : int
+ displayDetails() : void
+ calculateDepreciation() : double
+ getType() : string
Bike
-bikeType : string
-engineCC : int
+ displayDetails() : void
+ calculateDepreciation() : double
+ getType() : string
+ operator==(other:Bike) : bool
+ operator<<(os:ostream&, b:Bike) : ostream {friend}
Buyer
-favoriteCars[10] : Car
-favCount : int
+ displayInfo() : void
+ getRole() : string
+ addFavorite(c:Car) : void
+ removeFavorite(index:int) : void
+ displayFavorite() : void
+ sendMessage(m:Message*, s:string) : void
+ compareBuyers(a:buyer, b:buyer) : void {friend}
Seller
-sellerRating : double
-cars[10] : Car
-carCount : int
-sellerReview : Review
+ displayInfo() : void
+ getRole() : string
+ addCar(c:Car) : void
+ removeCar(index:int) : void
+ ShowReview() : void
+ getRating() : double
+ compareSellerRating(a:Seller, b:Seller) : void {friend}
Trusted Buyer
-trustBadge : string
+ displayInfo() : void
+ getRole() : string
Premium Seller
-badgeLevel : string
+ displayInfo() : void
+ getRole() : string
+ featureListing() : void
Admin
-role : string
-totalAdminUsers : int static
+ displayInfo() : void
+ getRole() : string
+ approveListing(v:IVehicle) : void
+ banSeller(s:Seller) : void
+ displayTotalAdmins() : void static
Obsi

Marketplace

- systemName : string
- allListing[50] : IVehicle*
- listingCount : int
- sellers[20] : Seller*
- sellerCount : int
- totalListings : static int
- totalUsers : static int
+ addUsersListing(v : IVehicle*) : void
+ deleteListing(index : int) : void
+ showAllListing() : void
+ searchByBrand(brand : string) : void
+ registerSeller(s : Seller*) : void
+ getListings() : IVehicle**
+ getListingCount() : int
+ getTotalListings() : int static

Date:
M T W T F S S

IUser {abstract}

- #userID : int
- #name: string
- # contact: string
- # email: string
- # displayInfo() void {virtual}
- # getRole(): string {virtual} {query}
- + login() void {virtual} {query}
- + getID() : int {query}
- + getName(): string {query}

IPayable {abstract}

- + processPayment(): void {virtual}
- + refundPayment(): void {virtual}
- + generateInvoice(): void {virtual}

IListable {abstract}

- + activateListing(): void {virtual}
- + deactivateListing(): void {virtual}
- + isActive(): bool {virtual} {query}
- + showListing(): void {query} {virtual}

IVehicle {abstract}

- + displayDetails(): void {query} {virtual}
- + calculateDepreciation(): double {query} {virtual}
- + getType(): string {virtual} {query}
- + getPrice(): double {virtual} {query}
- + getBrand(): string {virtual} {query}
- + getModel: string {virtual} {query}
- + getYear: int {virtual} {query}

Vehicle

- #brand, model: string
- # year : int
- # price: double
- # isSold: bool
- + operator=(v: Vehicle const): Vehicle
- + getBrand(): string {query}
- + getModel(): string {query}
- + getYear(): int {query}
- + getPrice(): double {query}
- + markAsSold(): void
- + operator<(other: Vehicle const): bool
- + operator>(other: Vehicle const): bool
- + operator==(other: Vehicle const): bool

Message

- - messageID, senderID, receiverID: const int
- - content, timestamp: string
- - status: bool
- + sendMessage(): void
- + markAsRead(): void
- + deleteMessage(): void
- + replyMessage(r: string): void

SearchFilter

- - brand, model: string
- - minPrice, maxPrice: double
- - minYear: int
- + matchVehicle(v: IVehicle) {query}: bool
- + applyFilter(size: int, vehicles: IVehicle*): void {query}
- + resetFilter(): void

Date:
M T W T F S

Payment
- paymentID: int
- amount: double
- buyerID, sellerID: int
- method: string
- paid: bool
- TAX-RATE: double const
+ processPayment(): void
+ refundPayment(): void
+ generateInvoice(): void
+ isPaid(): bool {query}
+ operator+(a:Payment, b:Payment) <<friend>>
+ ~~ostream~~ operator<<(os:ostream, p: Payment): ostream <<friend>>

Car
- mileage: float
- numDoors: int
- fuelType: string
+ operator=(c: Car): Car
+ displayDetails(): void
+ displayDetails(showFuel: bool): void
+ calculateDepreciation(): double
+ getType(): string
+ getMileage(): float
+ updateDetails(b: string, m: string, y: int, p: double, mil: float): void
+ operator+(other: Car): string
+ operator>>(is: istream, c: car): istream <<friend>>
+ operator>>(os: ostream, c: car): ostream <<friend>>

Listing
- listingID, views: int
- vehicle: IVehicle*
- datePosted: string
- active: bool
+ activateListing(): void
+ deactivateListing(): void
+ isActive(): bool {query}
+ showListing(): void {query}
+ incrementViews(): void

Electric Car
- batteryRangeKm: int
+ displayDetails(): void
+ calculateDepreciation(): double
+ getType(): string

Bike
- bikeType: string
- engineCC: int
+ displayDetails(): void
+ calculateDepreciation(): double
+ getType(): string
+ operator==(other: Bike): bool
+ operator<<(os: ostream, bk: Bike): ostream <<friend>>

Review
- reviewID, reviewerID, sellerID: int const
- rating: int
- comment, date: string
+ addReview(): void
+ deleteReview(): void
+ editReview(newC: string, newR: int): void
+ displayReview(): void
+ getRating(): int

User
- password: string
- isLoggedIn: bool
+ login(): void {query}
+ logout(): void
+ operator==(other: User): bool {query}

Buyer

- -favorites[10]: Car
- -favCount: int
- + displayInfo(): void
- + getRole(): string
- + addFavorite(c: Car): void
- + removeFavorite(index: int): void
- + displayFavorites(): void
- + sendMessage(m: ~~message~~ string): void
- + compareBuyers(a: buyer, b: buyer): void <<friends>>

Seller

- -sellerRating: double
- -cars[10]: Car
- -carCount: int
- -sellerReview: Review
- + displayInfo(): void
- + getRole(): string
- + addCar(c: Car): void
- + removeCar(index: int): void
- + showReview(): void
- + getRating(): double
- + compareSellerRating(a: Seller, b: Seller): void <<friends>>

Trusted Buyer

- -trustBadge: string
- + displayInfo(): void
- + getRole(): string

Premium Seller

- -badgeLevel: string
- + displayInfo(): void
- + getRole(): string
- + featureListing(): void

<<friends>>

Admin

- -role: string
- -totalAdmin: int static
- + displayInfo(): void
- + getRole(): string
- + approveListing(v: Vehicle): void
- + banSeller(s: Seller): void
- + displayTotalAdmins(): void static